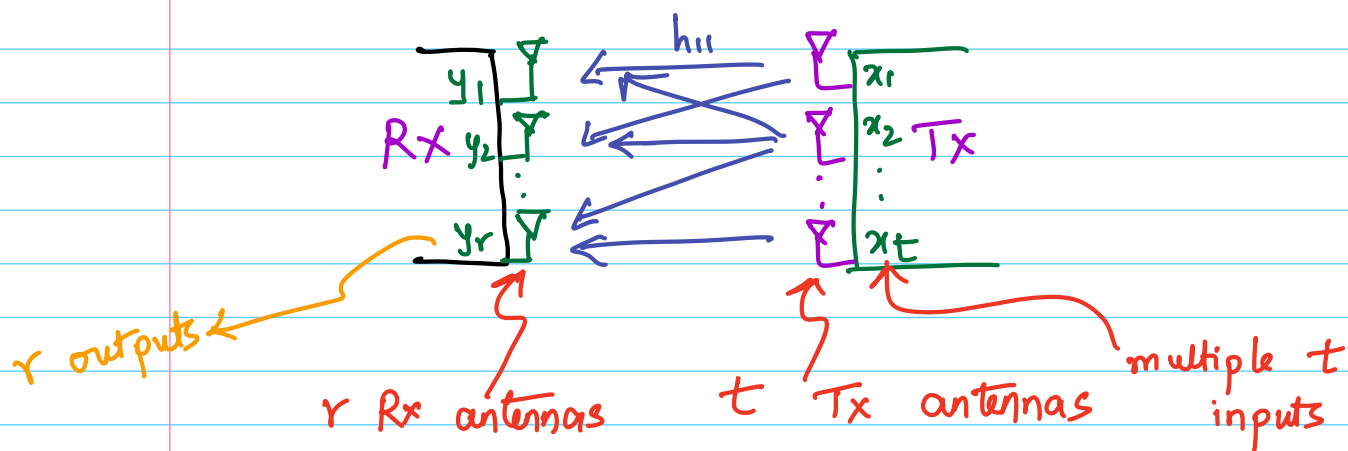


# Applications of Least Square

## MIMO RECEIVER



$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}}_{\underline{\mathbf{y}}} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1t} \\ h_{21} & h_{22} & \cdots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \cdots & h_{rt} \end{bmatrix}}_{\substack{r \times t \\ \text{MIMO channel} \\ \text{matrix}}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}}_{\underline{\mathbf{x}}} + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_r \end{bmatrix}}_{\underline{\mathbf{n}}}$$

$h_{ij}$  = channel coefficient  
btw Rx ant.  $i$   
by Tx ant.  $j$

$$\underline{\mathbf{y}} = \mathbf{H} \underline{\mathbf{x}} + \underline{\mathbf{n}}$$

MIMO system model

At the receiver how to determine  $\underline{\mathbf{x}}$ , given  $\underline{\mathbf{y}}$ .

if  $r = t$  &  $\mathbf{H}$  is invertible

$$\text{then } \hat{\underline{\mathbf{x}}} = \mathbf{H}^{-1} \underline{\mathbf{y}}$$

↑  
estimate of  $\underline{\mathbf{x}}$

Naturally, one cannot use this for  $r > t$

Therefore for  $r > t$   
determine  $\hat{x}$  as

$$\hat{x} = \arg \min \| \bar{y} - H \bar{x} \|^2$$

$$= (H^H H)^{-1} H^H \bar{y} = H^\dagger \bar{y}$$

$$\hat{x} = \underbrace{(H^H H)^{-1} H^H}_{\text{This is termed ZERO FORCING (ZF) Receiver}} \bar{y}$$

This is termed  
ZERO FORCING  
(ZF) Receiver

using Hermitian operator  
because  $H$  is typically  
complex

Because it forces  
interference between the  
symbols to zero.

$$\bar{y} = H \bar{x} + \bar{n}$$

$$y_i = \underbrace{h_{i1}x_1 + h_{i2}x_2 + \dots + h_{it}x_t}_{\text{All symbols } x_1, x_2, \dots, x_t \text{ are interfering at each Receive antenna } i.} + n_i$$

All symbols  $x_1, x_2, \dots, x_t$   
are interfering at each  
Receive antenna  $i$ .

$$\bar{y} = H \bar{x} + \bar{n}$$

Processed  
output  $\rightarrow \tilde{y}$

Apply the ZF receiver

$$\tilde{y} = (H^H H)^{-1} H^H \bar{y}$$

$$= (H^H H)^{-1} H^H (H \bar{x} + \bar{n})$$

$$= \bar{x} + \underbrace{H^T \bar{n}}_{\tilde{n}}$$

$$\tilde{y} = \bar{x} + \tilde{n}$$

$$\begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \tilde{y}_3 \\ \vdots \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix} + \begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \vdots \\ \tilde{n}_t \end{bmatrix}$$

$$\tilde{y}_i = \underbrace{x_i}_{\text{each } \tilde{y}_i \text{ depends only on } x_i} + \tilde{n}_i$$

$\Rightarrow$  interference = 0.

Therefore, this receiver forces interference to zero.

Hence it is termed as the ZF receiver.

Ex:

Consider the MIMO channel matrix

$$H = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

$\Rightarrow r = 4$  Rx ant.  
 $t = 2$  Tx ant.

$$H^T = (H^H H)^{-1} H^H$$

$$H^T H = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

$$(H^T H)^{-1} = \frac{1}{20} \begin{bmatrix} 30 & -10 \\ -10 & 4 \end{bmatrix}$$

$$(H^T H)^{-1} H^T = \frac{1}{20} \begin{bmatrix} 30 & -10 \\ -10 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

$2 \times 2$                        $2 \times 4$

$$= \frac{1}{20} \begin{bmatrix} 20 & 10 & 0 & -10 \\ -6 & -2 & 2 & 6 \end{bmatrix}$$

$$\therefore H^+ = \begin{bmatrix} 1 & 1/2 & 0 & -1/2 \\ -3/10 & -1/10 & 1/10 & 3/10 \end{bmatrix}$$

$$\begin{matrix} \hat{x} \\ 2 \times 1 \end{matrix} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = (H^T H)^{-1} H^T \underline{y}$$

$$= H^+ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

$4 \times 1$

$$\begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 & 0 & -1/2 \\ -3/10 & -1/10 & 1/10 & 3/10 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

$$\hat{x}_1 = y_1 + \frac{1}{2} y_2 + 0 \cdot y_3 - \frac{1}{2} y_4$$

$$\hat{x}_2 = -3/10 y_1 - 1/10 y_2 + 1/10 y_3 + 3/10 y_4$$

symbol estimates in the  $4 \times 2$  mimo system.

